**DSA06 DATA HANDLING AND VISUALIZATION | Activity: Identify Seasonality Patterns | Slot D**

**R.No: 16 Reg. No: 192424097 Name: K DEEKSHA SREE**

*Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; every student's series is different. Write your answers and paste your plots and code.*

## Series SP-D16: Retail sales [20 marks]

**Scenario:** You are the analyst for a retail chain. The series sales (Sales (lakh rupees)) is recorded quarterly. Management believes it follows a repeating pattern and wants it identified before planning. The cycle length is 4 observations. Run the code below to create the series x and answer the tasks.

**Evidence: run this code in R (do not edit it).**

```
set.seed(13848)
m <- 4                      # observations per cycle (quarterly data)
n <- m * 11                 # 11 full cycles
t <- 1:n
trend <- 125.2 + 0.952 * t
sales <- round(trend + 26.0 * cos(2 * pi * (t - 2) / m) + rnorm(n, 0, 6.9), 1)
x <- ts(sales, frequency = m, start = c(2015, 1))
```

**Tasks:**

1. Look at the series. Plot x (plot(x) or ggplot2). State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(sales ~ 1:n) and report the slope per quarter). [3]
2. The seasonal pattern. Use decompose(x) and extract $figure (the 4 seasonal values). Name the peak quarter and the trough quarter with their values, and the seasonal swing (peak minus trough). [4]
3. Is it consistent? Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 11 cycles is the highest value in the peak quarter from Task 2? Is the pattern consistent? [3]
4. Confirm with autocorrelation. Detrend (d <- resid(lm(sales ~ 1:n))) and compute acf(d, lag.max = 4, plot = FALSE) using observation lags. Report the autocorrelation at lag 4 and at lag 2, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality? [4]
5. Additive or multiplicative? Compute the range (max minus min) of sales in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify [2]. Remove the seasonal part with decompose(x, type = ...) using your chosen type (subtract for additive, divide for multiplicative), fit lm(adjusted ~ 1:n) and report the slope per quarter [2]. Write one sentence of advice to management based on your findings [2]. [6]

***Remember:*** *seasonality repeats at a fixed, known period; a trend is not a season; if the swings grow with the level, the season is multiplicative.*

*Code of Conduct: I certify that this work is my own. Signature: _____*

**1.Question:** *Plot x. State n, the cycle length, the number of full cycles, and whether the overall trend is rising, falling or flat (fit lm(sales ~ 1:n) and report the slope per quarter).*

## R code

```
n; m; n / m                          # 44, 4, 11

trend_fit <- lm(sales ~ I(1:n))
summary(trend_fit)$coefficients
round(coef(trend_fit)[2], 3)          # slope per quarter

df <- data.frame(time = as.numeric(time(x)), sales = sales)
ggplot(df, aes(time, sales)) +
  geom_line(colour = "#1F4E79") + geom_point(colour = "#1F4E79", size = 1.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              colour = "#B03A2E", linetype = "dashed") +
  labs(title = "Quarterly retail sales, 2015 to 2025 (Series SP-D16)",
       x = "Year", y = "Sales (lakh rupees)")
```

## Plot

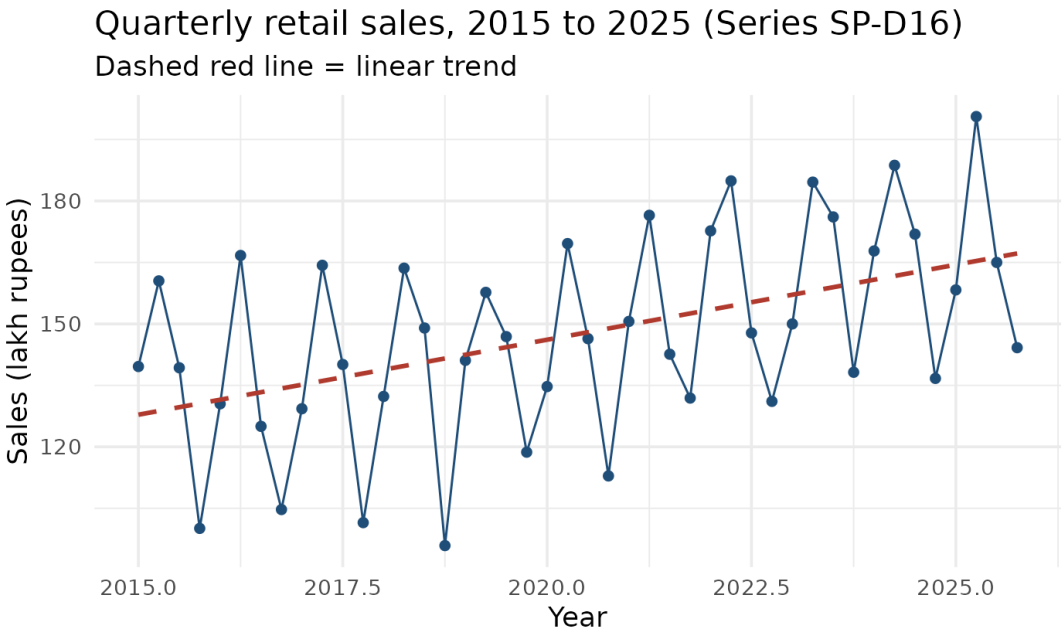



*Figure 1: Quarterly sales with a clear repeating wave around a slowly rising trend (dashed line).*

## Answer

| Quantity | Value |
|---|---|
| **n (observations)** | 44 |
| **Cycle length (m)** | 4 quarters |
| **Number of full cycles** | 11 (44 / 4) |
| **Slope of lm(sales ~ 1:n)** | +0.915 lakh rupees per quarter (p = 0.001) |
| **Overall trend** | Rising |

The overall trend is rising: sales increase by about 0.92 lakh rupees every quarter (about 3.7 lakh per year), and the series shows a regular up-and-down wave on top of this.

**2.Question:** *Use decompose(x) and extract $figure (the 4 seasonal values). Name the peak quarter and the trough quarter with their values, and the seasonal swing (peak minus trough).*

### R code

```
dc <- decompose(x)
round(dc$figure, 2)                 # seasonal values for Q1..Q4

peak   <- which.max(dc$figure); trough <- which.min(dc$figure)
peak; round(dc$figure[peak], 2)
trough; round(dc$figure[trough], 2)
round(max(dc$figure) - min(dc$figure), 2)   # seasonal swing

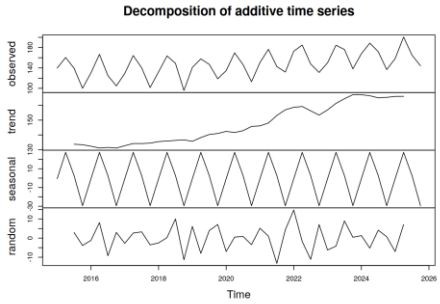
plot(dc)                            # observed, trend, seasonal, random
```

### Seasonal values

| Q1 | Q2 | Q3 | Q4 |
|---|---|---|---|
| -0.68 | 27.44 | 2.66 | -29.42 |

| Quantity | Value |
|---|---|
| **Peak quarter** | Q2 (April to June), seasonal value = +27.44 |
| **Trough quarter** | Q4 (October to December), seasonal value = -29.42 |
| **Seasonal swing (peak - trough)** | 27.44 - (-29.42) = 56.86 lakh rupees |

In a typical year, Q2 sales run about 27 lakh above the underlying level and Q4 sales about 29 lakh below it, while Q1 and Q3 stay close to the trend.




**3.Question:** *Draw a seasonal plot: one line per cycle against position in the cycle (ggplot2, group = cycle, colour = cycle; position = cycle(x)). In how many of the 11 cycles is the highest value in the peak quarter from Task 2? Is the pattern consistent?*

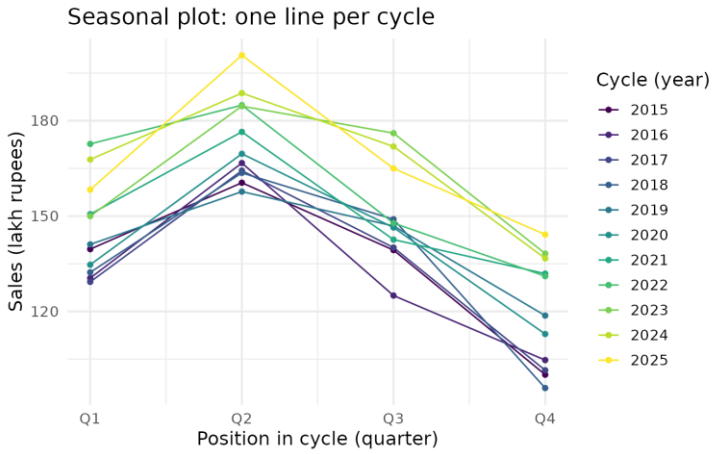
### R code

```
sd <- data.frame(
  cycle = factor(rep(1:11, each = 4)),
  pos   = as.numeric(cycle(x)),
  sales = as.numeric(x))

ggplot(sd, aes(pos, sales, group = cycle, colour = cycle)) +
  geom_line() + geom_point(size = 1.4) +
  scale_x_continuous(breaks = 1:4, labels = paste0("Q", 1:4)) +
  labs(title = "Seasonal plot: one line per cycle",
       x = "Position in cycle (quarter)", y = "Sales (lakh rupees)")

mat <- matrix(as.numeric(x), ncol = 4, byrow = TRUE)   # rows = cycles
best <- apply(mat, 1, which.max)                       # quarter of each cycle's maximum
best; sum(best == peak)
```

## Plot




**Highest value in the peak quarter:** In all 11 of the 11 cycles, the highest value of the year falls in Q2.

**Consistent?** Yes. Every cycle has the same shape (rise to Q2, ease at Q3, drop to the Q4 low), and the lines are roughly parallel and shift upward over the years because of the rising trend. The pattern is very consistent, which is the signature of true seasonality.

> **4.Question:** *Detrend (d <- resid(lm(sales ~ 1:n))) and compute acf(d, lag.max = 4, plot = FALSE) using observation lags. Report the autocorrelation at lag 4 and at lag 2, and compare with the significance bound 2/sqrt(n). What do the two values say about seasonality?*

## R code

```
d <- resid(lm(sales ~ I(1:n)))
a <- acf(d, lag.max = 4, plot = FALSE)
round(a$acf[, 1, 1], 2)            # lags 0, 1, 2, 3, 4

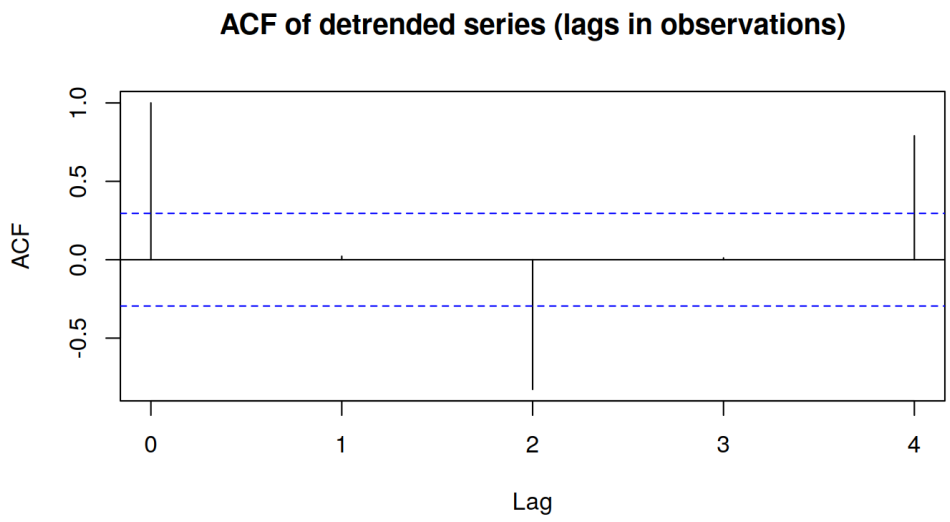
round(a$acf[5], 2)                 # lag 4
round(a$acf[3], 2)                 # lag 2
round(2 / sqrt(n), 2)              # significance bound

acf(d, lag.max = 4, main = "ACF of detrended series")
```

## Output

| Lag 0 | Lag 1 | Lag 2 | Lag 3 | Lag 4 |
|---|---|---|---|---|
| 1.00 | 0.02 | -0.83 | 0.01 | 0.79 |

| Quantity | Value |
|---|---|
| **Autocorrelation at lag 4** | +0.79 |
| **Autocorrelation at lag 2** | -0.83 |
| **Significance bound 2 / sqrt(n)** | 2 / sqrt(44) = 0.30 (about ±0.30) |




### Interpretation

Both values lie far outside the ±0.30 bound. The strong positive autocorrelation at lag 4 (0.79) means that a quarter resembles the same quarter one cycle earlier, which confirms a repeating pattern of length 4. The strong negative autocorrelation at lag 2 (-0.83) means that a quarter is the opposite of the quarter half a cycle away (a Q2 high pairs with a Q4 low). Together, with lags 1 and 3 near zero, they show a genuine seasonal cycle with period 4 quarters.

> **5.Question:** *Compute the range (max minus min) of sales in the first cycle and in the last cycle and their ratio (last / first); decide the type and justify. Remove the seasonal part with decompose(x, type = ...) using your chosen type and fit lm(adjusted ~ 1:n); report the slope per quarter. Write one sentence of advice to management based on your findings.*

### R code

```
first <- sales[1:4]; last <- sales[(n - 3):n]
r_first <- diff(range(first)); r_last <- diff(range(last))
r_first; r_last; round(r_last / r_first, 2)

dc_add   <- decompose(x, type = "additive")
adjusted <- x - dc_add$seasonal                 # additive: subtract
adj_fit  <- lm(as.numeric(adjusted) ~ I(1:n))
round(coef(adj_fit)[2], 2)                      # slope per quarter
```

### Part A: ranges

| Cycle | Values (Q1, Q2, Q3, Q4) | Range (max - min) |
|---|---|---|
| **First (2015)** | 139.6, 160.5, 139.3, 100.1 | 160.5 - 100.1 = 60.4 |
| **Last (2025)** | 158.3, 200.6, 165.0, 144.2 | 200.6 - 144.2 = 56.4 |
| **Ratio (last / first)** | | 56.4 / 60.4 = 0.93 |

**Type: ADDITIVE.** The ratio is about 0.93, close to 1, so the seasonal swing has stayed about the same size (roughly 56 to 60 lakh) even though the level of sales rose from an average of about 135 lakh in the first year to about 167 lakh in the last year. If the season were multiplicative, the swing would have grown with the level (a ratio well above 1); it did not.

### Part B: seasonally adjusted slope

| Quantity | Value |
|---|---|
| **Slope of lm(adjusted ~ 1:n), additive adjustment** | +1.00 lakh rupees per quarter |

| Quantity | Value |
|---|---|
| **For comparison: slope of the raw series (Task 1)** | +0.91 lakh rupees per quarter |

Removing the seasonal part gives a cleaner trend of about +1.0 lakh rupees per quarter (about 4 lakh per year), so the underlying growth is steady and slightly stronger than the raw fit suggests.

## Advice to management

Plan stock, staffing and budgets around a fixed yearly pattern of a Q2 peak (about 27 lakh above trend) and a Q4 low (about 29 lakh below trend), a constant swing of roughly 57 lakh that sits on top of steady growth of about 1 lakh per quarter, rather than reading each Q4 dip as a sign that the business is weakening.